

Carly Rodriguez, B.S.

Predoctoral Associate

The Jackson Laboratory (JAX)

600 East Main Street

Bar Harbor, ME 04609

carly.rodriguez@jax.org

EDUCATION:

- Ph.D.** Tufts Graduate School of Biomedical Sciences August 2023 - Present
Department of Neuroscience
PhD in Neuroscience, Neurogenetics Training Program at JAX
- B.S.** Ursinus College August 2019 – May 2023
Neuroscience
Minor in Spanish
Honors: Summa Cum Laude

RESEARCH EXPERIENCE:

Predoctoral Associate May 2024 – Present

The Jackson Laboratory, Bar Harbor, ME

Advisor: Dr. Martin F. Pera

- Executing original thesis research on genetic factors influencing neuronal resiliency to stress and aging using pluripotent stem cell models
- Mentored an undergraduate student in a 12-week summer program and academic year fellowship and continuing to serve as her senior honors project mentor
- Proficiency in techniques including high-content confocal microscopy on the Opera Phenix, immunocytochemistry, cortical neuron differentiation, data analysis in R studio, flow cytometry, digital PCR, and quantitative PCR

Undergraduate Researcher January 2020 – May 2023

Ursinus College, Collegeville, PA

Advisor: Dr. Jennifer King

- Worked on experiments investigating inflammatory response of microglia to HIV Tat protein *in vitro*
- Led independent honors project to establish a novel method at institution for the isolation of primary microglia from aged mice
- Trained and mentored new undergraduate students in the lab
- Gained proficiency in techniques including western blot, immunocytochemistry, cell culture maintenance, wide-field fluorescent imaging

Summer Research Intern

June 2022 – August 2022

University of California Irvine, Irvine, CA

Advisor: Dr. Lisa Flanagan

- Examined the impact of endothelial cell contact in mediating neural stem cell types
- Techniques include immunocytochemistry, cell culture maintenance, confocal fluorescent imaging, and FIJI Image Analysis

OTHER RELEVANT EXPERIENCE:

JAX Graduate Student Organization Events Coordinator

August 2025 – August 2026

The Jackson Laboratory, Bar Harbor, ME

- Provide logistical support for events, including making reservations, assessing availability, extending invitations and communications about events, and coordinating with necessary JAX committees and administrators for planning
- Help facilitate prospective JAX graduate student campus visitations and interviews

Molecular Neurobiology Laboratory Teaching Assistant

January 2023 – May 2023

Ursinus College, Collegeville, PA

Advisor: Dr. Jennifer King

- Aided in teaching and assisting undergraduate students with learning molecular laboratory techniques and completing their semester projects

Fundamentals of Neuroscience Teaching Assistant

January 2023 – May 2023

Ursinus College, Collegeville, PA

Advisor: Dr. Carlita Favero

- Provided assistance during lecture and laboratory activities for undergraduate students
- Aided in grading and providing feedback of course assignments
- Hosted exam review and tutoring sessions for students

PUBLICATIONS

- *In preparation*: Pera F. Martin, Rodriguez N. Carly. *Human embryonic stem cells*.
- Gutierrez B, Liu TC, **Rodriguez C**, et al. *Human endothelial cells promote a human neural stem cell type B phenotype via Notch signaling*. *Nat Commun*. 2025 May 30;16(1):5031. doi: 10.1038/s41467-025-60194-6.

PRESENTATIONS

Posters

- “Establishment of an *in vitro* platform to assess neuronal response to stress across genetic backgrounds” – Society for Neuroscience Annual Meeting, San Diego, CA, November 2025
- “Contact with human endothelial progenitor cells mediate a decrease in the proliferation of human neural stem and progenitor cells” – UCI Summer Institute in Neuroscience Symposium, August 2022

- “Time-course analysis of HIV-1 Tat protein exposure on microglial phagocytic activity” – Ursinus Celebration of Student Achievement, April 2022

Talks

- “Modeling the impact of genetic variation on age and stress in stem cell-derived cortical neurons” – Tufts Neuroscience Department Research Talks, February 2025 and JAX Trainee Talks, March 2025
- “Isolation of primary microglia from adult mice to examine neuroinflammatory response in neurodegenerative disease modeling”. Ursinus Celebration of Student Achievement, April 2023